U20CAM-4M UserManual



1. General Description:

The U20CAM-4M is a high-performance USB 2.0 UVC camera module designed for industrial vision, embedded systems, machine vision, intelligent terminals, and imaging applications. It is equipped with the OmniVision OV4688 4-megapixel CMOS image sensor, delivering high-resolution images with excellent color reproduction and stable performance.

Featuring native UVC (USB Video Class) compliance, the camera supports true plug-and-play operation without requiring additional drivers on Windows, Linux, macOS, Android, and other UVC-compatible operating systems. It provides both MJPEG and YUY2 video output formats, supporting multiple resolutions and high frame rates to meet a wide range of application requirements.

The OV4688 sensor integrates OmniBSI-2™ back-side illumination technology and HDR/WDR imaging, enabling outstanding image quality in both bright and low-light environments while maintaining excellent dynamic range.

With its compact 32x32 mm PCB design, standard USB interface, and flexible mounting holes, the U20CAM-4M is easy to integrate into various embedded and industrial devices. It is an ideal imaging solution for machine vision, robotics, AI applications, barcode recognition, industrial inspection, smart terminals, medical equipment, and other embedded vision systems.

2. Features

1. [Plug & Play UVC Camera] Native UVC compliant USB 2.0 camera module with driver-free setup for Windows, Mac OS, Linux, Ubuntu and Android.

2.[High-Frame-Rate] Support up to 4MP@30FPS, 2K@60FPS, 1080P@60FPS. Support YUY2 and MJPEG formats.

3.[True HDR/WDR Camera] Onboard OV4688 sensor features HDR and OmniBSI-2 back-illuminated technology, delivering clear and stable image quality even in low-light and challenging lighting environments.

4.[flexible Mounting & 71 degree]32x32mm size with 4x M2 holes. FOV(D)= 71 degree, FOV(H)= 58 degree

5.[Compliance And Safety] RoHS & REACH-SVHC compliant, tested by accredited labs

3. Technical Specification

Sensor	Type	OV4688
	Size	1/3 Inch
	Resolution	2688*1520
	Image Color	Colours
	Shutter type	Rolling shuttere
Lens	F(N) /Aperture	1.8
	Focal Length	3.96mm
	TV DISTORTION	＜1.0%
	CRA	＜34°
	Lens Diameter	M8.5*P0.25
	Lens Seat Spacing	18mm
	FOV-D	71°
	FOV-H	58°
PCBA	Size	32*32 mm
	Layers	4 layers
	Mounting Holes	4 counts, Diameter 2.2mm
	Trigger	None
	Strobe Flash	None
	Night vision	Reserved
Interfaces	Communication Interface	USB 2.0 Hgih-Speed
	Cable Length	1 M
	Device Name	U20CAM-4M
	VID/PID	0x364D/0X6203
	Communication Standard	UVC 1.0.0
	Supported format	YUY2, MJPEG
	Supported Frame rate	30 fps

Software	Supported Resolution	MJPEG Output resolutions: 2688*1520 @30fps 2560*1440 @60fps 1920*1080 @60fps 1600*1200 @60fps 1280x960 @60fps 1280x720 @60fps 1024x768 @60fps 960x540 @60fps 800x600 @60fps 640x480 @60fps 352x288 @60fps 320x240 @60fps 160x120 @60fps YUY2 Output resolutions: 2688*1520 @1fps 1920*1080 @3fps 1600*1200 @3fps 1280x960 @6fps 1280x720 @8fps 1024x768 @8fps 960x540 @8fps 800x600 @18fps 640x480 @30fps 352x288 @30fps 320x240 @30fps 160x120 @30fps
	Auto Parameters	White Balance（Manual Option） Exposure（Manual Option） Auto Focus （Manual Option）
	Controllable Parameters	Brightness, Contrast, Hue, Saturation, Sharpness, Gamma, White Balance, Backlight Comp, Gain, Exposure, PowerLine Frequency, Roll, Low Light Compensation
	Support System	Windows , Linux , Mac Os, Android

4. User Manual Version Descriptions

Version	Description	Date	E-mail
V1.0		2026.05.21	support@inno-maker.com sales@inno-maker.com

If you have any suggestions, ideas, codes and tools please feel free to email to me. I will update the user manual and record your name and E-mail in list. Look forward to your letter and kindly share.